
When Does Inference Routing Break Down? Disentangling Temperature Artifacts from Domain-Dependent Scaling Limits

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Abstract

Inference routers that select between depth scaling (chain-of-thought) and width scaling (self-consistency) assume the optimal strategy is a stable, predictable property of each problem. We demonstrate that this assumption is undermined by a pervasive experimental confound: temperature asymmetry between strategies. Under standard protocols where chain-of-thought uses lower temperature than self-consistency, three of four evaluation domains exhibit an apparent strategy reversal at high token budgets. Chain-of-thought dominates at 1k tokens, but self-consistency overtakes at 16k. However, matched-temperature experiments (both strategies at $\tau=0.8$, three seeds per domain, McNemar’s test) reveal that this reversal is largely artifactual: on competition mathematics and GPQA Diamond, the two strategies are statistically indistinguishable once temperature is controlled ($p > 0.77$ on all seeds). A third domain, multi-hop QA, shows the same non-significant pattern, but near-floor accuracy ($\sim 17\%$) limits statistical power ($p > 0.32$). Only MMLU-Pro retains a genuine self-consistency advantage (+4.8 to +8.3 percentage points, significant on two of three seeds at $p < 0.01$, directionally consistent in the third at $p = 0.09$). A sample-efficiency analysis explains this divergence: on knowledge-intensive tasks, even a single self-consistency sample outperforms chain-of-thought, whereas on reasoning-heavy tasks, all $k=4$ samples are needed merely to match it, at $4\times$ the compute cost for zero gain. The mechanism is a depth scaling ceiling: 44–70% of chain-of-thought outputs at 16k tokens enter repetition loops, and effective reasoning content (median 680–1,230 words) varies across domain types. Our findings reframe inference routing breakdown from a budget-dependent phenomenon to a domain-dependent one, and expose temperature confounding as a systematic threat to the routing literature.

1 Introduction

Inference-time compute scaling has become a primary lever for improving large language model (LLM) reasoning without increasing model parameters. Snell et al. [2024] showed that optimally allocating test-time compute can outperform scaling model size. Their work establishes a compute-optimal frontier whose shape depends on problem difficulty. Two canonical strategies sit at opposite ends of this frontier: *depth scaling* invests all tokens in a single extended chain-of-thought (CoT) trace [Wei et al., 2022], while *width scaling* distributes the same budget across multiple independent samples aggregated by majority vote (self-consistency, SC) [Wang et al., 2023]. Because neither strategy uniformly dominates, a growing line of work proposes *inference routers*, lightweight classifiers that predict, per instance, the strategy most likely to yield the correct answer [Snell et al., 2024, Ong et al., 2024]. These routers rest on an implicit assumption: that the optimal strategy is a *stable, per-instance property*. In this view, some problems inherently favor deeper reasoning, others benefit from broader sampling, and this preference persists as compute scales. Despite the practical

importance of this assumption, no prior work has tested it by varying the compute budget while holding the model fixed.

This assumption appears to fail under standard evaluation protocols. On 975 problems across four domains (competition mathematics [Hendrycks et al., 2021], GPQA Diamond [Rein et al., 2023], MMLU-Pro [Wang et al., 2024], and multi-hop QA) evaluated at 1k, 4k, and 16k token budgets on Qwen3-32B, CoT is uniformly preferred at 1k tokens and wins 67–87% of discordant pairs (Table 4). Yet at 16k tokens, SC@4 overtakes CoT on three of four domains by 7–10 percentage points (Table 1). The reversal is stark on competition math: CoT dominates at 1k (23.6% vs. 5.5%) but falls behind SC@4 at 16k (58.6% vs. 65.8%). Taken at face value, this pattern suggests routing is fundamentally fragile. A router trained at low budgets would not merely become uninformative at higher budgets but would *actively misroute*, and would produce outcomes worse than always selecting the globally dominant strategy.

There is, however, a confound hiding in plain sight. Standard protocols assign different sampling temperatures to the two strategies ($\tau=0.6$ for CoT, $\tau=0.8$ for SC), in line with common practice but with the effect of introducing a systematic asymmetry that has not been scrutinized in the routing literature. Temperature modulates both the quality of sequential reasoning (higher τ can suppress the repetition loops in extended CoT traces) and the diversity of parallel samples (higher τ increases voting-relevant variation in SC). Any comparison across strategies at mismatched temperatures therefore conflates the strategy effect with the temperature effect. This is not a minor calibration issue: temperature determines whether a 16k CoT trace produces over a thousand words of novel reasoning or degenerates into repetitive loops, a difference that can swing accuracy by double digits. Crucially, the confound’s direction and magnitude are domain-dependent. Open-ended reasoning tasks, where CoT traces are long enough to enter repetition, are far more sensitive to temperature than structured tasks with short answers.

Matched-temperature experiments separate the strategy signal from the temperature artifact. We evaluate both strategies at $\tau=0.8$ and 16k tokens across three random seeds and test each comparison with McNemar’s exact test [McNemar, 1947] (Table 2). The result is striking: three of four apparent reversals vanish. On competition math, the strategies achieve near-identical accuracy (71.1% vs. 70.8%; $p \in [0.91, 1.00]$ across seeds). The *entire* 7.2 pp gap under the standard protocol is a temperature artifact. On GPQA Diamond, the strategies are similarly indistinguishable (57.1% vs. 56.7%; $p \in [0.77, 0.90]$), with the sign of the gap flipping across seeds. Only MMLU-Pro retains a genuine SC advantage (+6.6 pp on average, $p < 0.01$ on two of three seeds), a result that reflects majority-vote denoising over structured multiple-choice answers rather than any temperature effect. This finding reframes routing breakdown as *domain-dependent*, not budget-dependent: the question is not “at what budget does routing fail?” but “for which domains does depth-vs-width carry a real signal once confounds are removed?” A sample-efficiency analysis (SC@ k) illuminates the mechanism: on MMLU-Pro, even a single width sample outperforms full-budget CoT; on reasoning-intensive domains, $k=4$ samples merely match CoT, at $4\times$ the compute cost for zero net gain. A depth scaling ceiling compounds the picture: 44–70% of 16k CoT outputs enter repetition loops and cap effective reasoning at 680–1,230 words, but the ceiling’s impact on routing viability is itself domain-dependent. The practical implication is immediate: routing systems that rely on unmatched-temperature comparisons to determine strategy preferences will misroute on the majority of domains.

We make three contributions:

1. We identify **temperature confounding** as a systematic threat to inference routing evaluation. Under matched temperature ($\tau=0.8$, three seeds, McNemar’s test), three of four apparent strategy reversals at 16k tokens are artifactual ($p > 0.77$ on competition math and GPQA; $p > 0.32$ on multi-hop QA, which is underpowered at near-floor accuracy). Any routing policy derived from unmatched-temperature comparisons risks active misrouting on the majority of domains.
2. We **reframe routing breakdown as domain-dependent**. Only knowledge-intensive tasks (MMLU-Pro) exhibit a genuine SC advantage after temperature control (+6.6 pp on average). Reasoning-intensive domains show no residual strategy difference. The relevant axis for routing feasibility is the task’s knowledge demands and answer structure, not the token budget.
3. We provide a **mechanistic analysis** that links domain type to routing outcome. Sample-efficiency curves (SC@ k) show that knowledge tasks benefit from width at any k , while reasoning tasks

require $k \geq 4$ for zero gain. A depth scaling ceiling and a majority-vote failure mode on free-text QA (82% all-unique answers, oracle 28.8% vs. voted 17.2%) complete the picture.

2 Related Work

2.1 Inference-Time Compute Scaling

Chain-of-thought (CoT) prompting [Wei et al., 2022] demonstrated that intermediate reasoning steps substantially improve LLM performance on complex tasks. Zero-shot variants [Kojima et al., 2022] further showed that simple instructions such as “think step by step” suffice to elicit such reasoning. Self-consistency [Wang et al., 2023] showed that sampling multiple reasoning paths and aggregating via majority voting further improves accuracy, and established the canonical depth-vs-width tradeoff we study. Fu et al. [2023] extended this line by showing that selecting more complex reasoning chains as prompts improves multi-step reasoning. Brown et al. [2024] demonstrated that repeated sampling scales coverage on code and math benchmarks, with $\text{pass}@k$ increasing log-linearly in the number of samples.

Snell et al. [2024] provided a foundational analysis of compute-optimal inference. They showed that the effectiveness of different test-time strategies varies with problem difficulty and that adaptive compute allocation can outperform uniform scaling. Our work is complementary: while Snell et al. study how to allocate test-time compute given a fixed strategy space, we study how the *comparability* of strategies changes when experimental confounds (temperature asymmetry) are controlled. This analysis reveals that much of the apparent strategy variation may be artifactual.

Li et al. [2024] demonstrated that majority-vote performance scales with the number of sampled agents. Our sample-efficiency analysis qualifies this finding: the scaling benefit of width (more samples) is domain-dependent. On knowledge-intensive tasks, even a single sample suffices; on reasoning-intensive tasks, four samples merely match single-trace CoT.

2.2 Inference Routing and Strategy Selection

The idea of routing queries to different models or strategies has been explored primarily in the multi-model setting. RouteLLM [Ong et al., 2024] trains routers using preference data to dynamically select between a stronger and weaker LLM. RouterBench [Hu et al., 2024] provides a benchmark for evaluating multi-LLM routing systems. Chen et al. [2023] proposed cascading and routing strategies to reduce LLM inference costs. Shnitzer et al. [2023] and Lu et al. [2024] further developed benchmark-based and reward-guided routing methods. These approaches route across *models* of different capability; we instead route across *strategies* applied to a single model.

Our work differs from the routing literature in a fundamental way: we do not propose a new router. Instead, we characterize the conditions under which depth-vs-width routing is *feasible*, and we show that the routing signal observed under standard protocols is largely a temperature artifact. To our knowledge, no prior work has examined whether reported strategy differences in inference routing survive temperature matching.

2.3 Depth vs. Width in Reasoning

Best-of- n sampling with learned verifiers [Cobbe et al., 2021, Lightman et al., 2024] generates multiple candidate solutions and selects the highest-scoring one. Tree of Thoughts [Yao et al., 2023] combines depth and width by exploring multiple reasoning branches with lookahead. Process reward models [Lightman et al., 2024] provide step-level verification for finer-grained selection among reasoning paths. These approaches represent richer strategy spaces than the binary CoT-vs-SC choice we study. Our temperature-confounding finding applies in principle to any comparison between strategies evaluated at different temperatures.

2.4 Repetition and Degeneration in LLMs

Neural text degeneration, where model outputs become repetitive or incoherent, is well-documented. Holtzman et al. [2020] showed that maximization-based decoding leads to repetitive loops and proposed nucleus sampling as a mitigation. Welleck et al. [2020] addressed repetition through unlike-

likelihood training. Fu et al. [2021] provided a theoretical analysis of the repetition problem. Xu et al. [2022] proposed methods to detect and mitigate repetitive generation.

Wu et al. [2026] showed that task accuracy follows an inverted U-shaped curve with CoT length, and that the optimal length scales with task difficulty. Our finding is complementary: we identify repetition loops as a generation pathology that prevents models from reaching even a task-appropriate reasoning length and creates a hard ceiling below the optimal depth. At 16k tokens, 44–70% of CoT outputs enter repetition loops with median effective content of only 680–1,230 words.

2.5 Temperature and Sampling in LLMs

Temperature scaling, originally introduced in the context of knowledge distillation [Hinton et al., 2015], modulates the entropy of the output distribution. Renze and Guven [2024] studied the effect of sampling temperature on LLM problem-solving across benchmarks and found that the optimal temperature varies by task. Alternative sampling strategies such as top- k [Fan et al., 2018] and locally typical sampling [Meister et al., 2023] address similar tradeoffs between diversity and quality. Our work highlights a specific consequence of temperature choice: when different strategies are evaluated at different temperatures, the resulting confound can dominate the measured strategy difference.

3 Method

3.1 Problem Formulation

We formalize inference routing as a budget-conditioned strategy selection problem. Let $\mathcal{X}$ denote the space of problem instances and $\mathcal{Y}$ the answer space. An *inference strategy* $s \in \mathcal{S}$ maps an instance x and token budget b to a distribution over answers. We consider two strategies: **Depth (CoT)** allocates the full budget b to a single chain-of-thought trace; **Width (SC@ k)** distributes b across k independent samples (b/k tokens each) and aggregates via majority voting [Wang et al., 2023]. We use $k=4$. An *inference router* $r : \mathcal{X} \times \mathcal{B} \rightarrow \mathcal{S}$ selects the strategy predicted to yield the correct answer. We treat b as a free variable to study how routing viability changes across the scaling curve.

3.2 Standard-Protocol Experiments

Model and sampling. All experiments use Qwen3-32B [Yang et al., 2024] with FP8 quantization, served via vLLM [Kwon et al., 2023]. Following conventional practice, CoT uses temperature $\tau=0.6$ and SC@4 uses $\tau=0.8$ (to encourage sample diversity for majority voting), with $p=1.0$ (no nucleus truncation) and no top- k filtering. All generations use zero-shot chain-of-thought prompting [Kojima et al., 2022].

Budget grid. $b \in \{1k, 4k, 16k\}$ tokens, spanning over an order of magnitude. At 1k, SC@4 is limited to 250 tokens per sample; at 16k, each SC sample receives 4k tokens, above the typical reasoning saturation point observed in our analysis (Section 4.5).

Evaluation domains. Four domains ($N=975$ problems): competition mathematics ($n=275$) from MATH [Hendrycks et al., 2021], graduate-level science from GPQA Diamond ($n=198$) [Rein et al., 2023], professional knowledge from MMLU-Pro ($n=252$) [Wang et al., 2024], and multi-hop QA ($n=250$) based on HotpotQA [Yang et al., 2018]. These domains span a spectrum from structured multiple-choice (MMLU-Pro) to open-ended free-text (multi-hop QA) and enable analysis of how answer format interacts with strategy choice. Each problem is evaluated under both strategies at each budget (5,850 inference runs).

Answer extraction and scoring. For competition math, we extract the final boxed expression and evaluate via symbolic equivalence. For GPQA Diamond and MMLU-Pro (multiple-choice), we extract the selected option letter. For multi-hop QA, we use token-level $F1 \geq 0.5$ for single-trace evaluation. SC majority voting uses exact string matching to identify the plurality answer across the k samples; ties are broken randomly. For multi-hop QA, we replace exact-string majority voting with normalized-match voting following SQuAD conventions [Rajpurkar et al., 2016]: answers are

lowercased, stripped of articles and punctuation, and two answers are considered equivalent if their token-level F1 ≥ 0.8 .

Deduplication at 16k. At 16k, 44–70% of CoT outputs enter repetition loops where the model cycles through similar reasoning without progress. We apply deterministic deduplication: six parallel detectors operating at multiple granularities (character n -gram, sentence, and paragraph level) flag repetitive segments, and the trace is truncated at the earliest flagged offset (sentence-aligned). The answer is re-extracted from the truncated trace. All 16k CoT numbers under the standard protocol use deduplication-corrected values (CoT_{dedup}); Appendix D analyzes side effects, including a net -7.6 pp impact on competition math from aggressive truncation.

3.3 Matched-Temperature Control Experiments

The standard protocol assigns $\tau=0.6$ to CoT and $\tau=0.8$ to SC. This temperature asymmetry is a potential confound: observed strategy differences may reflect temperature effects on generation quality rather than genuine depth-vs-width tradeoffs. Higher temperature suppresses the repetition loops in extended CoT traces while simultaneously increasing sample diversity for SC voting. Both effects favor the strategy already assigned the higher temperature.

Design. Both strategies are evaluated at $\tau=0.8$ with 16k tokens, the setting where the standard protocol shows the most pronounced reversals. We choose $\tau=0.8$ as the matched temperature because (1) matching at $\tau=0.6$ would disadvantage SC by reducing sample diversity, and (2) $\tau=0.8$ represents the best case for both strategies under their respective mechanisms. Each domain is evaluated under three independent seeds (42, 123, 456), yielding 24 additional inference configurations.

Statistical testing. For each domain–seed pair, we apply McNemar’s exact test [McNemar, 1947] to the 2×2 per-instance correctness contingency table. McNemar’s test is appropriate because both strategies evaluate the same instances; it conditions on discordant pairs only (instances where exactly one strategy is correct) and provides a paired test of marginal homogeneity with exact binomial p -values.

3.4 Sample-Efficiency Analysis

To understand why SC@4 outperforms CoT on some domains but not others, we simulate SC@ k for $k \in \{1, 2, 3, 4\}$ by drawing k of 4 available samples without replacement, applying majority voting, and averaging over 20 random subsamples per problem. This decomposes the SC advantage into a *per-sample component* (SC@1 vs. CoT) and a *voting component* (SC@4 vs. SC@1), and reveals whether width provides immediate per-sample gains or only marginal aggregation benefits.

3.5 Regime Classification

For each domain d and budget b , let n_{CoT} and n_{SC} denote the number of discordant pairs where only CoT or only SC is correct. The regime accuracy is:

$$\text{RegimeAcc}(d, b) = \frac{\max(n_{\text{CoT}}, n_{\text{SC}})}{n_{\text{CoT}} + n_{\text{SC}}}. \quad (1)$$

We classify each pair as *predictive* ($\geq 65\%$), *ambiguous* (55–65%), or *non-predictive* ($< 55\%$), with 95% bootstrap CIs [Efron and Tibshirani, 1993] ($B=10,000$). Full regime results are in Appendix B.

3.6 Router

We train a Tier-1 router (logistic regression on text surface features: lexical statistics, structural indicators, and domain flags) to predict the correct strategy, evaluated via 5-fold cross-validation. Baselines: a *majority baseline* (always selects the globally better strategy at each budget) and a *random baseline* (50%). Full results in Appendix C.

Table 1: Strategy accuracy (%) under standard protocol ($\tau_{\text{CoT}}=0.6$, $\tau_{\text{SC}}=0.8$). $\text{CoT}_{\text{dedup}}$ at 16k denotes deduplication-corrected accuracy. Under this protocol, SC@4 overtakes CoT at 16k on three of four domains, an apparent reversal that motivates matched-temperature analysis (Section 4.2).

Domain	1k tokens		4k tokens		16k tokens	
	CoT	SC@4	CoT	SC@4	$\text{CoT}_{\text{dedup}}$	SC@4
Competition Math	23.6	5.5	56.0	34.5	58.6	65.8
GPQA Diamond	35.9	21.7	51.0	46.5	49.0	56.6
MMLU-Pro	56.8	30.6	60.3	62.7	58.3	67.9
Multi-Hop QA	21.6	14.4	23.6	20.0	20.4	18.8

Table 2: Matched-temperature strategy comparison ($\tau=0.8$ for both strategies, 16k tokens). Three of four domains show no significant difference once temperature is controlled. Only MMLU-Pro retains a genuine SC@4 advantage.

Domain	CoT (%)	SC@4 (%)	Δ	p (range)	Verdict	Mean across 3 seeds
Competition Math	71.1	70.8	-0.3	0.91-1.00	No difference	
GPQA Diamond	57.1	56.7	-0.3	0.77-0.90	No difference	
MMLU-Pro	60.6	67.2	+6.6	0.0001-0.09	SC wins	
Multi-Hop QA	16.9	18.4	+1.5	0.32-0.85	No difference	

(42, 123, 456). $\Delta = \text{SC@4} - \text{CoT}$. McNemar’s exact test, two-sided. Per-seed details in Appendix A.

4 Results

4.1 Standard Protocol: An Apparent Strategy Reversal

Table 1 presents accuracy under the standard protocol ($\tau_{\text{CoT}}=0.6$, $\tau_{\text{SC}}=0.8$). Three patterns emerge across the budget axis.

CoT dominates at low budgets. At 1k tokens, CoT outperforms SC@4 on every domain: +18.1 pp on competition math, +14.2 pp on GPQA Diamond, +26.2 pp on MMLU-Pro, and +7.2 pp on multi-hop QA. With only 250 tokens per sample, SC@4 cannot generate sufficiently developed reasoning for any individual trace to be reliable, and majority voting over four weak samples does not compensate.

Strategy preferences begin to shift at moderate budgets. At 4k tokens, MMLU-Pro becomes the first domain where SC@4 overtakes CoT (62.7% vs. 60.3%). The remaining three domains still favor CoT, but margins narrow on GPQA Diamond (from +14.2 to +4.5 pp) and multi-hop QA (from +7.2 to +3.6 pp). Competition math is the exception: CoT’s absolute margin widens at 4k (+21.5 pp) as CoT scales steeply from 23.6% to 56.0%.

Direction reversal at high budgets. At 16k tokens, three of four domains flip to SC@4-dominant: competition math (65.8% vs. $\text{CoT}_{\text{dedup}}$ 58.6%, +7.2 pp), GPQA Diamond (56.6% vs. 49.0%, +7.6 pp), and MMLU-Pro (67.9% vs. 58.3%, +9.6 pp). Only multi-hop QA retains a slim CoT edge (20.4% vs. 18.8%), though both strategies perform near floor. Regime analysis (Appendix B) confirms that the routing signal is *predictive* at 1k (67-87% CoT majority among discordant pairs) but transitions to ambiguous or reversed at higher budgets. Taken at face value, this constitutes a routing breakdown: a router trained on low-budget data would actively select the wrong strategy at high budgets.

4.2 Temperature Confounding: Most Reversals Are Artifactual

Table 2 presents matched-temperature results ($\tau=0.8$ for both strategies, 16k tokens, three seeds).

Three of four reversals vanish. On competition mathematics, the strategies achieve near-identical accuracy (71.1% vs. 70.8%; $p = 0.91-1.00$ across seeds, with discordant pairs splitting nearly evenly). On GPQA Diamond, the strategies are similarly indistinguishable (57.1% vs. 56.7%;

$p = 0.77\text{--}0.90$), with the sign of the gap flipping across seeds (Appendix A). On multi-hop QA, three-seed CoT accuracy averages $16.9 \pm 1.7\%$ (seed 42: 15.6%, seed 123: 16.4%, seed 456: 18.8%; registry `exp_matched_temp_multihop_seed{42,123,456}`). These numbers confirm near-floor performance with good cross-seed stability. Under normalized-match SC voting (Section 3.2), SC@4 averages $18.4 \pm 0.4\%$ across three seeds (seed 42: 18.4%, seed 123: 18.8%, seed 456: 18.0%). The resulting $\Delta = +1.5$ pp on average with no seed reaching significance (McNemar p range 0.32–0.85, Fisher combined $p = 0.646$). The direction is inconsistent across seeds (+2.8, +2.4, -0.8 pp), and confirms a statistical tie. The standard-protocol reversals on these three domains were driven entirely by temperature asymmetry.

Mechanism: temperature rescues depth scaling. The confound operates asymmetrically across strategies. Raising CoT temperature from 0.6 to 0.8 suppresses repetition loops in extended traces, rescuing depth scaling: competition math CoT gains +12.5 pp (from 58.6% to 71.1%) while SC@4 gains only +5.0 pp (65.8% to 70.8%). On GPQA Diamond, the pattern is similar: CoT gains +8.1 pp vs. SC’s +0.1 pp. The gap that appeared to be a depth-vs-width difference was entirely explained by temperature’s differential effect on repetition-prone long traces versus short independent samples.

MMLU-Pro: a genuine domain-level effect. SC@4 outperforms CoT by +6.6 pp on average ($p < 0.01$ on two of three seeds, $p = 0.0001$ and $p = 0.008$; the third trends in the same direction, $p = 0.09$; per-seed details in Appendix A), a robust advantage that persists under matched temperature. Temperature has negligible effect on either MMLU-Pro strategy ($\sim +1.2$ pp each). This confirms that the advantage reflects majority-vote denoising over structured multiple-choice answers rather than any temperature artifact. The standard-protocol reversal on MMLU-Pro was real; temperature confounding merely inflated its magnitude.

4.3 Sample Efficiency Explains the Domain Split

Why does MMLU-Pro show a genuine SC advantage while the other three domains do not? SC@ k analysis ($k \in \{1, 2, 3, 4\}$, via subsampling without replacement, 20 random draws per problem) reveals a sharp dichotomy.

On **MMLU-Pro**, even SC@1 outperforms CoT. The advantage is inherent in per-sample accuracy, not dependent on voting aggregation. Each independent sample has a higher probability of selecting the correct answer than the single CoT trace, likely because short structured answers are less susceptible to the reasoning drift that affects extended traces. Additional samples compound this advantage through majority-vote denoising: SC@2 adds ~ 3 pp over SC@1, and SC@4 adds another ~ 2 pp.

On **competition math**, **GPQA**, and **multi-hop QA**, SC@1 performs *below* CoT. Accuracy improves gradually with k , but at $k=4$, SC merely matches CoT, at $4\times$ the compute cost for zero net gain. This reflects a fundamental task-type distinction: knowledge-intensive tasks exhibit *independent* errors across samples (each may retrieve or apply different knowledge), so voting is effective. Reasoning-intensive tasks exhibit *correlated* errors: if the model lacks the reasoning capacity for a problem, multiple samples fail in similar ways, and voting is ineffective.

4.4 Majority-Vote Failure on Free-Text QA

Multi-hop QA exposes a structural failure mode of majority voting on free-text answers. Under exact string matching, open-ended questions elicit diverse surface forms even when the underlying answer is correct: at 16k tokens ($\tau=0.8$), 82% of SC@4 instances produce four unique answer strings with no plurality winner, forcing a random tie-break. Replacing exact matching with normalized-match voting following SQuAD conventions (Section 3.2) partially mitigates this: voted accuracy rises from 17.2% (exact) to 18.0% (normalized; seed 456, $n=250$; registry `exp_matched_temp_multihop_seed456`). However, oracle accuracy remains 28.8%, and normalized voting still *destroys* over a third of the signal that width scaling produces (oracle gap = 10.8 pp). Across three seeds, SC_{norm} averages 18.4% while SC_{exact} (seeds 42 and 456) averages $\sim 17.4\%$; normalization recovers only ~ 1 pp, far from closing the oracle gap. Near-floor accuracy limits the interpretability of this gap; results indicate that answer-space fragmentation is a structural property of free-text SC aggregation rather than a remediable engineering choice. This failure is

domain-specific: structured tasks with canonical answer forms (MMLU-Pro, competition math) do not exhibit it.

4.5 Depth Scaling Ceiling

At 16k tokens, 44% (competition math) to 70% (multi-hop QA) of CoT outputs enter repetition loops.¹ Median effective token count before repetition onset: 1,100 (competition math), 808 (GPQA), 790 (MMLU-Pro), 481 (multi-hop QA). The model’s reasoning capacity saturates at 3–7% of the allocated budget.

SC@4 at 16k allocates 4,000 tokens per sample, above the typical reasoning saturation point, so individual samples complete without entering loops. On MMLU-Pro, SC’s per-sample advantage *compounds* with the ceiling: width wins both because individual samples are better and because CoT depth is wasted. On reasoning-intensive domains, the ceiling affects both strategies symmetrically under matched temperature: CoT wastes tokens on repetition and SC wastes them on redundant reasoning paths, so the net result is equivalence rather than SC dominance. The matched-temperature controls reveal that the ceiling’s routing impact is itself domain-dependent: it drives a genuine SC advantage only on knowledge-intensive tasks where per-sample accuracy already favors width.

5 Discussion

Temperature confounding as a systematic threat. The most consequential finding is methodological. Assigning different temperatures to different strategies is widespread and well-motivated: lower entropy aids coherent reasoning, higher entropy aids diverse sampling [Renze and Guven, 2024]. Yet this practice introduces a confound that can dominate the measured strategy effect. On competition math, the standard protocol shows a 7.2 pp SC advantage at 16k, while matched-temperature experiments show zero difference. The confound operates through an asymmetric mechanism: higher temperature suppresses the repetition loops in extended CoT traces, and confers a large accuracy gain on depth scaling (+12.5 pp on competition math) while having a much smaller effect on the shorter SC samples (+5.0 pp). Any reported strategy comparison conducted under mismatched temperatures should be verified with matched controls before being used to calibrate routing policies.

Domain-dependent routing: mechanistic explanation. Once temperature is controlled, the landscape simplifies along a single axis: answer structure. Knowledge-intensive tasks with short structured answers (MMLU-Pro) exhibit *independent* per-sample errors. Each sample may retrieve or weight different knowledge, and majority-vote denoising is effective at any sample count. Reasoning-intensive tasks (competition math, GPQA) exhibit *correlated* errors. If the model cannot solve a problem, multiple samples fail in the same way, and voting provides no correction. This explains why SC@1 already outperforms CoT on MMLU-Pro but SC@4 merely matches CoT on reasoning domains: the error correlation structure, not the number of samples, determines whether width scaling pays off. Domain-level routing (always use SC on knowledge tasks) is trivial and captures all real performance differences; per-instance routing within a domain provides no additional value.

Depth scaling ceiling and voting failure. Repetition-induced saturation at 481–1,100 effective tokens establishes a practical ceiling on depth scaling. Any system that relies on extended single-trace reasoning may encounter this ceiling, though our matched-temperature results show that temperature tuning largely offsets its accuracy impact in reasoning-intensive domains. Conversely, multi-hop QA reveals a ceiling on width scaling: exact-match majority voting over free-text answers destroys over a third of the oracle signal (Section 4.4). Richer aggregation methods (e.g., token F1-based voting or learned verifiers [Lightman et al., 2024]) are needed for open-ended tasks.

Implications for system design. Three directions follow. First, *domain-level routing should replace instance-level routing*: a simple domain classifier captures all real strategy differences. Second, *temperature should be treated as a routing variable*: the strategy×temperature interaction is more consequential than strategy choice alone. Third, *budget-aware stopping* can convert wasted

¹Rates under matched temperature ($\tau=0.8$). Standard-protocol rates ($\tau=0.6$) are similar: 44–70%.

depth tokens into useful breadth by detecting reasoning saturation via repetition monitoring and reallocating remaining budget to width samples.

6 Limitations

All experiments use a single model, Qwen3-32B [Yang et al., 2024] with FP8 quantization, and the depth scaling ceiling and repetition loops may be model-specific. Generalizing the findings requires replication across architectures and scales. We match temperature at $\tau=0.8$ only; matching at other temperatures might yield different conclusions, particularly for SC where lower temperature reduces sample diversity. The matched-temperature analysis covers only the 16k budget, so we have not verified whether the temperature confound explains the signal decay at 4k or whether the 1k regime is robust to temperature matching.

We evaluate only CoT and SC@4. Other values of k , alternative aggregation (weighted voting, best-of- n with verifiers [Lightman et al., 2024]), or hybrid strategies may exhibit different dynamics. On competition math, deduplication flips 23 outputs from correct to incorrect vs. 2 in reverse (−7.6 pp net), though under matched temperature this is moot because competition math shows no strategy difference regardless.

Broader Impact Statement

This work is a methodological study of inference-time compute allocation for large language models. We identify a temperature confound in existing evaluation protocols and show that routing between depth and width scaling strategies is domain-dependent rather than budget-dependent. Our findings may improve the efficiency of LLM deployment by guiding practitioners toward more reliable evaluation practices. We do not foresee negative societal impacts specific to this work beyond those inherent to LLM research more broadly.

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A Per-Seed Matched-Temperature Results

Table 3 presents the full per-seed matched-temperature results ($\tau=0.8$, 16k tokens).

Table 3: Per-seed matched-temperature results. McNemar’s exact test, two-sided.

Domain	Seed	CoT (%)	SC@4 (%)	p	Verdict
Competition Math	123	70.4	70.4	0.905	No difference
Competition Math	456	70.2	70.2	1.000	No difference
GPQA Diamond	42	57.6	56.1	0.784	No difference
GPQA Diamond	123	56.6	58.1	0.771	No difference
GPQA Diamond	456	57.1	56.1	0.896	No difference
MMLU-Pro	123	59.5	67.9	<0.001	SC wins
MMLU-Pro	456	61.5	68.3	0.008	SC wins
Multi-Hop QA	456	18.8	17.2	0.572	No difference

B Routing Regime Classification

Table 4 maps each domain–budget pair to a routing regime under the standard protocol, based on the discordant-pair majority proportion (Eq. 1). All four domains are *predictive* at 1k (67–87% CoT majority). At 4k, three transition to *ambiguous*. At 16k, three domains show a direction flip (SC@4 dominant), while multi-hop QA decays to *non-predictive*. As shown in Section 4.2, the direction flips at 16k are largely explained by temperature confounding.

Table 4: Routing regime classification under the standard protocol ($\tau_{\text{CoT}}=0.6$, $\tau_{\text{SC}}=0.8$). Bootstrap CI from $B=10,000$ resamples.

Domain	1k tokens	4k tokens	16k tokens
Competition Math	86.8% [77.9, 94.1]	76.6% [68.5, 84.7]	61.6% [52.3, 72.1] [†]
GPQA Diamond	66.7% [56.0, 76.2]	57.4% [50.8, 70.5]	64.2% [50.9, 77.4] [†]
MMLU-Pro	79.5% [71.4, 86.6]	56.8% [50.0, 70.5]	78.6% [66.7, 90.5] [†]
Multi-Hop QA	72.5% [57.5, 85.0]	62.2% [51.3, 78.4]	54.0% [50.0, 68.0]
Predictive ($\geq 65\%$) Ambiguous (55–65%) Non-predictive ($< 55\%$)			
[†] Direction flip: SC@4 dominates at 16k (reversed from CoT at 1k).			

C Router Performance

Table 5 evaluates the Tier-1 router (logistic regression on text surface features) under the standard protocol. The router achieves AUC of 0.537 (1k) and 0.569 (4k), underperforming the majority baseline by 7.9–24.0pp. The matched-temperature analysis (Section 4.2) provides a deeper explanation: on three of four domains at 16k, there is no routing signal to capture because the strategies are equivalent once temperature is controlled.

Table 5: Tier-1 router performance under the standard protocol.

Method	1k	4k	AUC (1k / 4k)
Majority baseline	76.6%	64.0%	—
Tier-1 router (LR)	52.6%	56.1%	0.537 / 0.569
Random baseline	50.0%	50.0%	0.50 / 0.50

D Deduplication Analysis

The deduplication procedure (Section 3.2) truncates CoT traces at the earliest detected repetition onset, which is not symmetric: on competition math, 23 outputs flip from correct to incorrect vs. only 2 in reverse, yielding a net 7.6pp decrease (66.2% $\rightarrow$ 58.6%). The competition math reversal under the standard protocol depends on this correction; GPQA Diamond and MMLU-Pro reversals are dedup-independent. Under matched temperature, this is moot: competition math shows no strategy difference regardless of deduplication.